



1. Description

1.1. Project

Project Name	WireStripper
Board Name	NUCLEO-F446ZE
Generated with:	STM32CubeMX 6.16.0
Date	01/02/2026

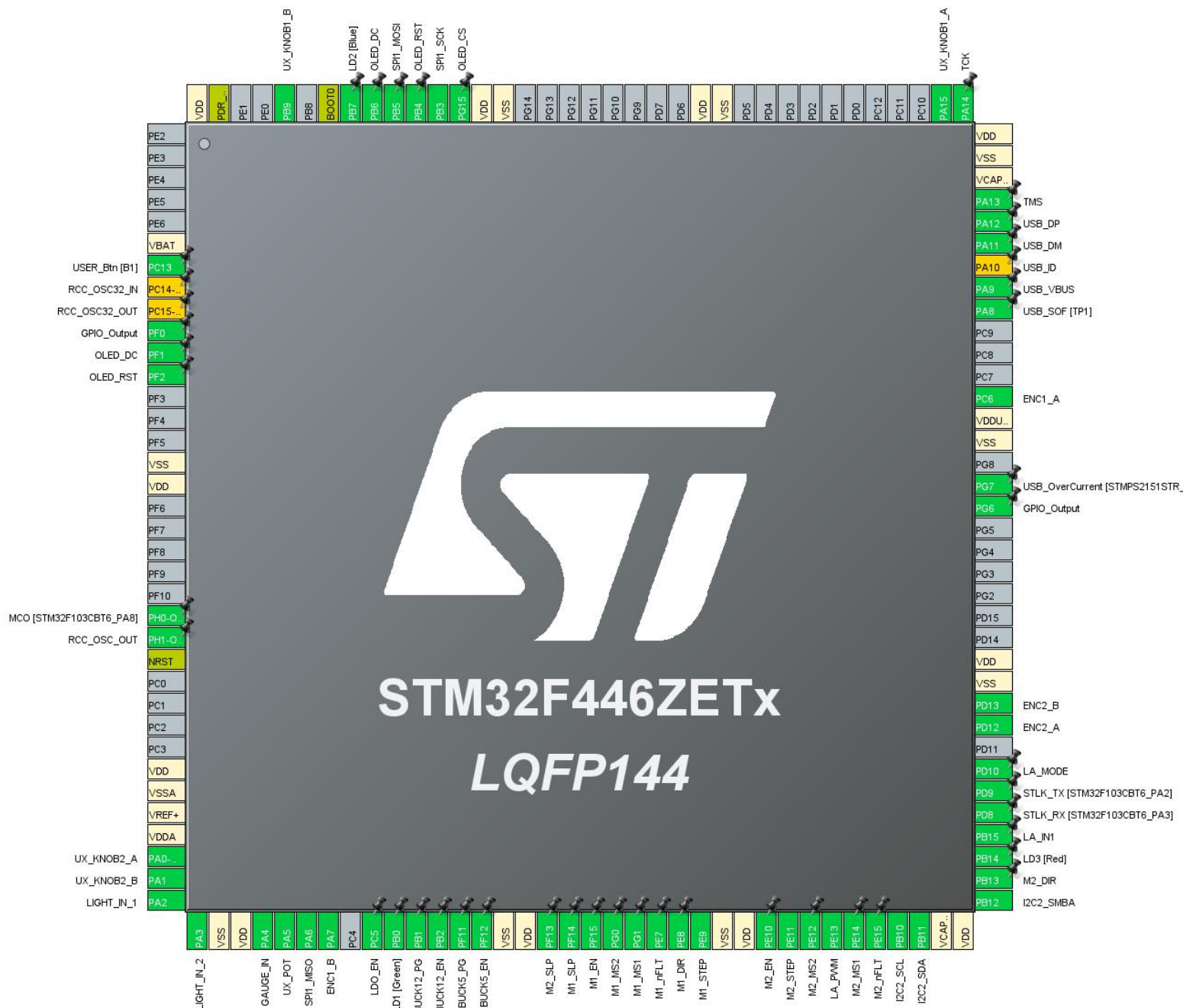
1.2. MCU

MCU Series	STM32F4
MCU Line	STM32F446
MCU name	STM32F446ZETx
MCU Package	LQFP144
MCU Pin number	144

1.3. Core(s) information

Core(s)	Arm Cortex-M4
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2. Pinout Configuration



3. Pins Configuration

Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
6	VBAT	Power		
7	PC13	I/O	GPIO_EXTI13	USER_Btn [B1]
8	PC14-OSC32_IN *	I/O	RCC_OSC32_IN	
9	PC15-OSC32_OUT *	I/O	RCC_OSC32_OUT	
10	PF0 **	I/O	GPIO_Output	
11	PF1 **	I/O	GPIO_Output	OLED_DC
12	PF2 **	I/O	GPIO_Output	OLED_RST
16	VSS	Power		
17	VDD	Power		
23	PH0-OSC_IN	I/O	RCC_OSC_IN	MCO [STM32F103CBT6_PA8]
24	PH1-OSC_OUT	I/O	RCC_OSC_OUT	
25	NRST	Reset		
30	VDD	Power		
31	VSSA	Power		
32	VREF+	Power		
33	VDDA	Power		
34	PA0-WKUP	I/O	TIM5_CH1	UX_KNOB2_A
35	PA1	I/O	TIM5_CH2	UX_KNOB2_B
36	PA2	I/O	ADC1_IN2	LIGHT_IN_1
37	PA3	I/O	ADC1_IN3	LIGHT_IN_2
38	VSS	Power		
39	VDD	Power		
40	PA4	I/O	ADC1_IN4	GAUGE_IN
41	PA5	I/O	ADC1_IN5	UX_POT
42	PA6	I/O	SPI1_MISO	
43	PA7	I/O	TIM3_CH2	ENC1_B
45	PC5 **	I/O	GPIO_Output	LDO_EN
46	PB0 **	I/O	GPIO_Output	LD1 [Green]
47	PB1 **	I/O	GPIO_Input	BUCK12_PG
48	PB2 **	I/O	GPIO_Output	BUCK12_EN
49	PF11 **	I/O	GPIO_Input	BUCK5_PG
50	PF12 **	I/O	GPIO_Output	BUCK5_EN
51	VSS	Power		
52	VDD	Power		
53	PF13 **	I/O	GPIO_Output	M2_SLP

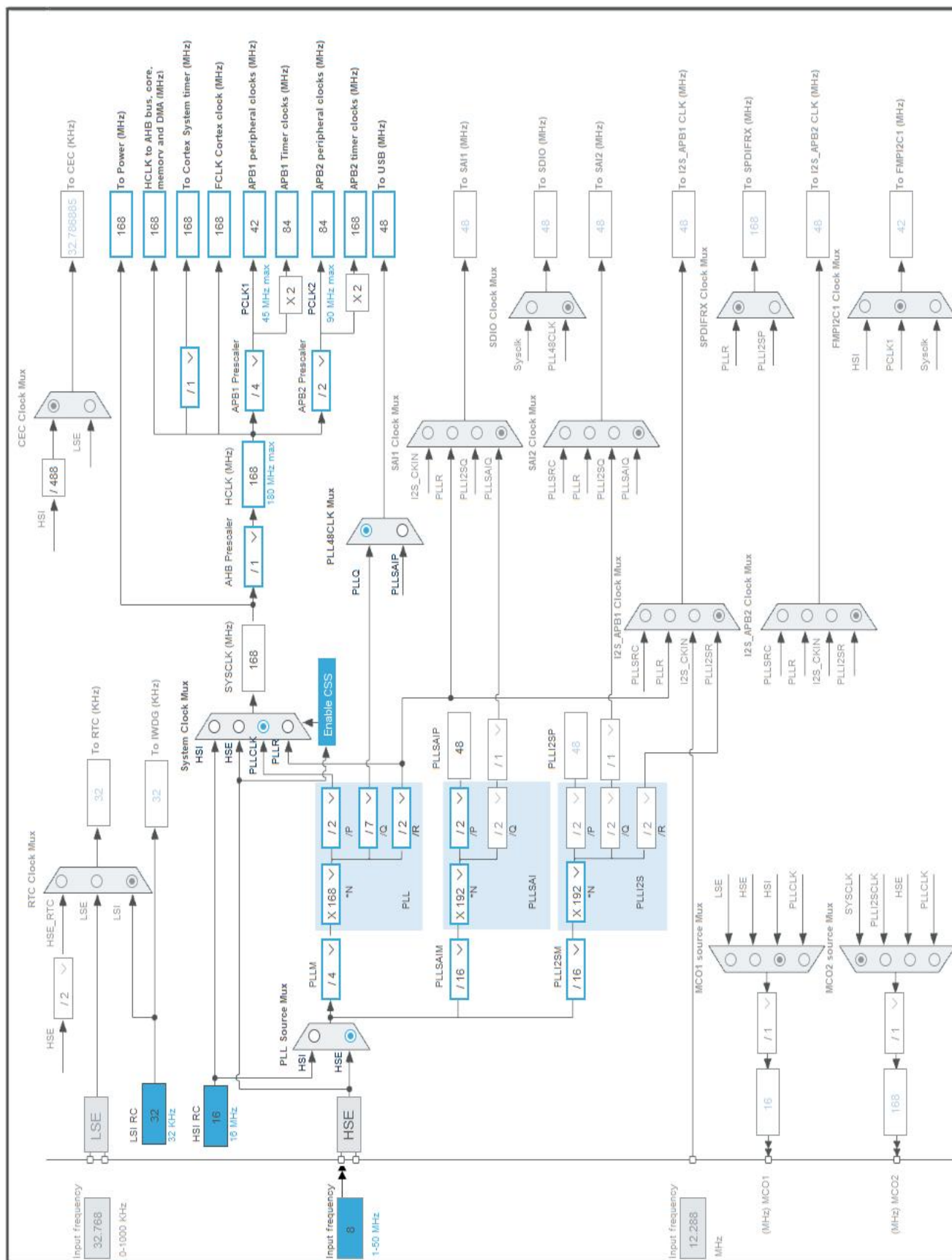
Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
54	PF14 **	I/O	GPIO_Output	M1_SLP
55	PF15 **	I/O	GPIO_Output	M1_EN
56	PG0 **	I/O	GPIO_Output	M1_MS2
57	PG1 **	I/O	GPIO_Output	M1_MS1
58	PE7 **	I/O	GPIO_Output	M1_nFLT
59	PE8 **	I/O	GPIO_Output	M1_DIR
60	PE9	I/O	TIM1_CH1	M1_STEP
61	VSS	Power		
62	VDD	Power		
63	PE10 **	I/O	GPIO_Output	M2_EN
64	PE11	I/O	TIM1_CH2	M2_STEP
65	PE12 **	I/O	GPIO_Output	M2_MS2
66	PE13	I/O	TIM1_CH3	LA_PWM
67	PE14 **	I/O	GPIO_Output	M2_MS1
68	PE15 **	I/O	GPIO_Output	M2_nFLT
69	PB10	I/O	I2C2_SCL	
70	PB11	I/O	I2C2_SDA	
71	VCAP_1	Power		
72	VDD	Power		
73	PB12	I/O	I2C2_SMBA	
74	PB13 **	I/O	GPIO_Output	M2_DIR
75	PB14 **	I/O	GPIO_Output	LD3 [Red]
76	PB15 **	I/O	GPIO_Output	LA_IN1
77	PD8	I/O	USART3_TX	STLK_RX [STM32F103CBT6_PA3]
78	PD9	I/O	USART3_RX	STLK_TX [STM32F103CBT6_PA2]
79	PD10 **	I/O	GPIO_Output	LA_MODE
81	PD12	I/O	TIM4_CH1	ENC2_A
82	PD13	I/O	TIM4_CH2	ENC2_B
83	VSS	Power		
84	VDD	Power		
91	PG6 **	I/O	GPIO_Output	
92	PG7 **	I/O	GPIO_Input	USB_OverCurrent [STMP2151STR_FAULT]
94	VSS	Power		
95	VDDUSB	Power		
96	PC6	I/O	TIM3_CH1	ENC1_A
100	PA8	I/O	USB_OTG_FS_SOF	USB_SOF [TP1]
101	PA9	I/O	USB_OTG_FS_VBUS	USB_VBUS

Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
102	PA10 *	I/O	USB_OTG_FS_ID	USB_ID
103	PA11	I/O	USB_OTG_FS_DM	USB_DM
104	PA12	I/O	USB_OTG_FS_DP	USB_DP
105	PA13	I/O	SYS_JTMS-SWDIO	TMS
106	VCAP_2	Power		
107	VSS	Power		
108	VDD	Power		
109	PA14	I/O	SYS_JTCK-SWCLK	TCK
110	PA15	I/O	TIM2_CH1	UX_KNOB1_A
120	VSS	Power		
121	VDD	Power		
130	VSS	Power		
131	VDD	Power		
132	PG15 **	I/O	GPIO_Output	OLED_CS
133	PB3	I/O	SPI1_SCK	
134	PB4 **	I/O	GPIO_Output	OLED_RST
135	PB5	I/O	SPI1_MOSI	
136	PB6 **	I/O	GPIO_Output	OLED_DC
137	PB7 **	I/O	GPIO_Output	LD2 [Blue]
138	BOOT0	Boot		
140	PB9	I/O	TIM2_CH2	UX_KNOB1_B
143	PDR_ON	Reset		
144	VDD	Power		

** The pin is affected with an I/O function

* The pin is affected with a peripheral function but no peripheral mode is activated

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32F4
Line	STM32F446
MCU	STM32F446ZETx
Datasheet	DS10693 Rev6

1.2. Parameter Selection

Temperature	25
Vdd	3.3

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

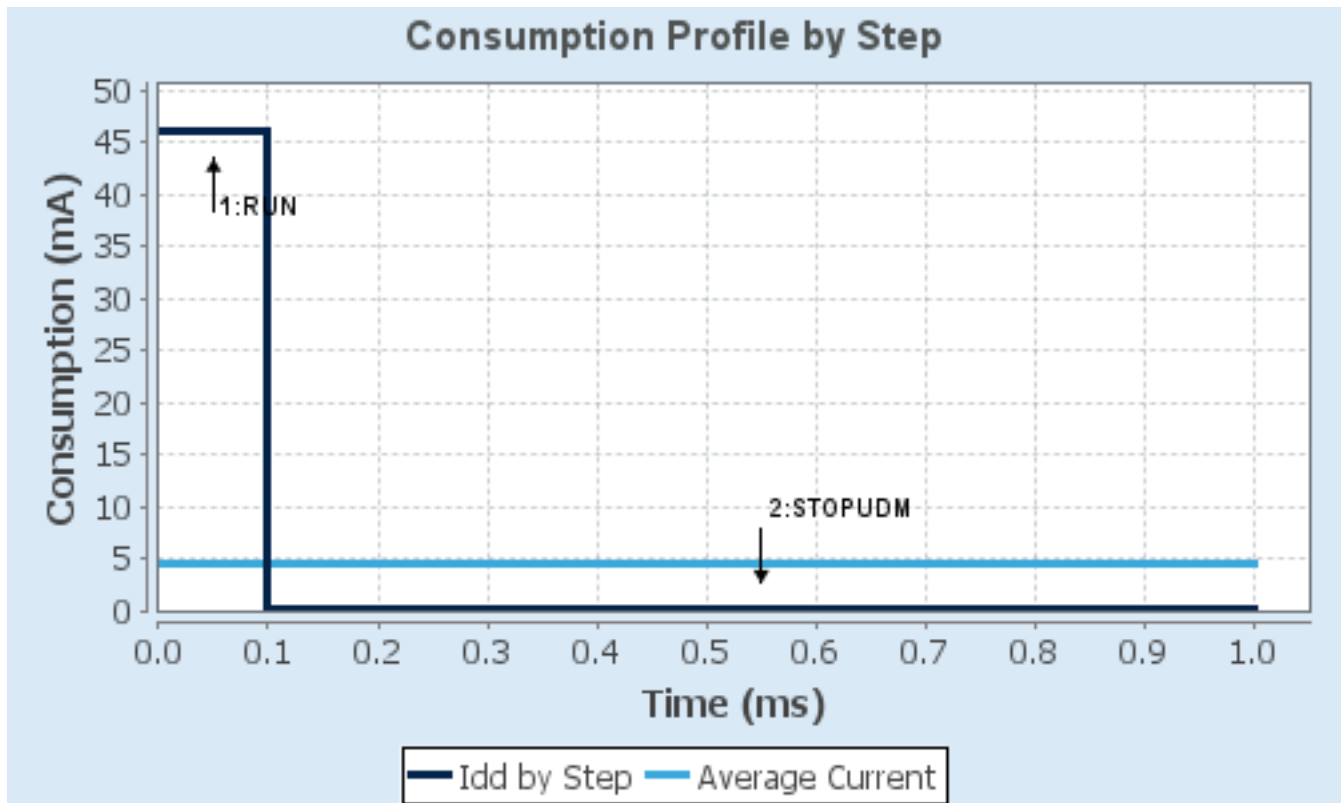
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP_UDM (Under Drive)
Vdd	3.3	3.3
Voltage Source	Battery	Battery
Range	Scale1-High	No Scale
Fetch Type	RAM/FLASH/REGON/ART/P REFETCH	n/a
CPU Frequency	180 MHz	0 Hz
Clock Configuration	HSE PLL	Regulator_LP Flash-PwrDwn
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	46 mA	55 μ A
Duration	0.1 ms	0.9 ms
DMIPS	225.0	0.0
Ta Max	99.99	104.99
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	4.65 mA
Battery Life	1 month	Average DMIPS	225.0 DMIPS

1.6. Chart



2. Software Project

2.1. Project Settings

Name	Value
Project Name	WireStripper
Project Folder	C:\Users\Admin\Desktop\4A\MTTE481\strip-master\WireStripper
Toolchain / IDE	CMake
Firmware Package Name and Version	STM32Cube FW_F4 V1.28.3
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x1000
Minimum Stack Size	0x1000

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	Yes
Backup previously generated files when re-generating	Yes
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	Yes
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_USART3_UART_Init	USART3
4	MX_ADC1_Init	ADC1
5	MX_SPI1_Init	SPI1
6	MX_TIM1_Init	TIM1
7	MX_TIM2_Init	TIM2
8	MX_TIM3_Init	TIM3
9	MX_TIM4_Init	TIM4
10	MX_TIM5_Init	TIM5
11	MX_USB_DEVICE_Init	USB_DEVICE

Rank	Function Name	Peripheral Instance Name
12	MX_I2C2_SMBUS_Init	I2C2

3. Peripherals and Middlewares Configuration

3.1. ADC1

mode: IN2

mode: IN3

mode: IN4

mode: IN5

3.1.1. Parameter Settings:

ADCs_Common_Settings:

Mode	Independent mode
ADC_Settings:	
Clock Prescaler	PCLK2 divided by 4
Resolution	12 bits (15 ADC Clock cycles)
Data Alignment	Right alignment
Scan Conversion Mode	Disabled
Continuous Conversion Mode	Disabled
Discontinuous Conversion Mode	Disabled
DMA Continuous Requests	Disabled
End Of Conversion Selection	EOC flag at the end of single channel conversion

ADC_Regular_ConversionMode:

Number Of Conversion	1
External Trigger Conversion Source	Regular Conversion launched by software
External Trigger Conversion Edge	None
<u>Rank</u>	1
Channel	Channel 2
Sampling Time	3 Cycles

ADC_Injected_ConversionMode:

Number Of Conversions	0
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WatchDog:

Enable Analog WatchDog Mode	false
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3.2. I2C2

I2C: SMBus-Alert-mode

3.2.1. Parameter Settings:

Timing configuration:

Clock Speed (Hz)	100000
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SMBus Slave Features:

Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
General Call Address Detection	Disabled
Clock No Stretch Mode	Disabled

SMBus Features:

Packet Error Check Mode	PEC Disabled
Peripheral Mode	Peripheral Mode Smbus Slave

Timeout configuration:

Timeout	0x00008019
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3.3. RCC

High Speed Clock (HSE): BYPASS Clock Source

3.3.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Enabled
Data Cache	Enabled
Flash Latency(WS)	5 WS (6 CPU cycle)

RCC Parameters:

HSI Calibration Value	16
TIM Prescaler Selection	Disabled
HSE Startup Timeout Value (ms)	100
LSE Startup Timeout Value (ms)	5000

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 1
Power Over Drive	Disabled

3.4. SPI1

Mode: Full-Duplex Master

3.4.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits

First Bit	MSB First
Clock Parameters:	
Prescaler (for Baud Rate)	2
Baud Rate	42.0 MBits/s *
Clock Polarity (CPOL)	High *
Clock Phase (CPHA)	2 Edge *
Advanced Parameters:	
CRC Calculation	Disabled
NSS Signal Type	Software

3.5. SYS

Debug: Serial Wire

Timebase Source: TIM6

3.6. TIM1

Channel1: PWM Generation CH1

Channel2: PWM Generation CH2

Channel3: PWM Generation CH3

3.6.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
Repetition Counter (RCR - 8 bits value)	0
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Break And Dead Time management - BRK Configuration:

BRK State	Disable
BRK Polarity	High

Break And Dead Time management - Output Configuration:

Automatic Output State	Disable
Off State Selection for Run Mode (OSSR)	Disable
Off State Selection for Idle Mode (OSSI)	Disable
Lock Configuration	Off

PWM Generation Channel 1:

Mode	PWM mode 1
Pulse (16 bits value)	20000 *
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

PWM Generation Channel 2:

Mode	PWM mode 1
Pulse (16 bits value)	20000 *
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

PWM Generation Channel 3:

Mode	PWM mode 1
Pulse (16 bits value)	20000 *
Output compare preload	Enable
Fast Mode	Disable
CH Polarity	High
CH Idle State	Reset

3.7. TIM2

Combined Channels: Encoder Mode

3.7.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode	Encoder Mode TI1
____ Parameters for Channel 1 ____	
Polarity	Rising Edge

IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0
____ Parameters for Channel 2 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

3.8. TIM3

Combined Channels: Encoder Mode

3.8.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode	Encoder Mode T11
____ Parameters for Channel 1 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0
____ Parameters for Channel 2 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

3.9. TIM4

Combined Channels: Encoder Mode

3.9.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	65535
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode	Encoder Mode TI1
____ Parameters for Channel 1 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0
____ Parameters for Channel 2 ____	
Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

3.10. TIM5

Combined Channels: Encoder Mode

3.10.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection	Reset (UG bit from TIMx_EGR)

Encoder:

Encoder Mode	Encoder Mode TI1
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____ Parameters for Channel 1 ____

Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

____ Parameters for Channel 2 ____

Polarity	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter	0

3.11. USART3

Mode: Asynchronous

3.11.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples

3.12. USB_OTG_FS

Mode: Device_Only

mode: Activate_SOF

mode: Activate_VBUS

3.12.1. Parameter Settings:

Speed	Device Full Speed 12MBit/s
Low power	Disabled
Link Power Management	Disabled
VBUS sensing	Enabled
Signal start of frame	Enabled

3.13. FREERTOS

Interface: CMSIS_V2

3.13.1. Config parameters:

API:

FreeRTOS API CMSIS v2

Versions:

FreeRTOS version 10.3.1

CMSIS-RTOS version 2.1.3

MPU/FPU:

ENABLE_MPU Disabled

ENABLE_FPU Disabled

Kernel settings:

USE_PREEMPTION Enabled

CPU_CLOCK_HZ SystemCoreClock

TICK_RATE_HZ 1000

MAX_PRIORITIES 56

USE_SB_COMPLETED_CALLBACK 0

USE_MINI_LIST_ITEM 1

MINIMAL_STACK_SIZE **256 ***

MAX_TASK_NAME_LEN 16

USE_16_BIT_TICKS Disabled

IDLE_SHOULD_YIELD Enabled

USE_MUTEXES Enabled

USE_RECURSIVE_MUTEXES Enabled

USE_COUNTING_SEMAPHORES Enabled

QUEUE_REGISTRY_SIZE 8

USE_APPLICATION_TASK_TAG Disabled

ENABLE_BACKWARD_COMPATIBILITY Enabled

USE_PORT_OPTIMISED_TASK_SELECTION Disabled

USE_TICKLESS_IDLE Disabled

USE_TASK_NOTIFICATIONS Enabled

RECORD_STACK_HIGH_ADDRESS **Enabled ***

Memory management settings:

Memory Allocation Dynamic / Static

TOTAL_HEAP_SIZE **50000 ***

Memory Management scheme heap_4

Hook function related definitions:

USE_IDLE_HOOK Disabled

USE_TICK_HOOK	Disabled
USE_MALLOC_FAILED_HOOK	Enabled *
USE_DAEMON_TASK_STARTUP_HOOK	Disabled
CHECK_FOR_STACK_OVERFLOW	Option2 *

Run time and task stats gathering related definitions:

GENERATE_RUN_TIME_STATS	Disabled
USE_TRACE_FACILITY	Enabled
USE_STATS_FORMATTING_FUNCTIONS	Disabled

Co-routine related definitions:

USE_CO_ROUTINES	Disabled
MAX_CO_ROUTINE_PRIORITIES	2

Software timer definitions:

USE_TIMERS	Enabled
TIMER_TASK_PRIORITY	2
TIMER_QUEUE_LENGTH	10
TIMER_TASK_STACK_DEPTH	512

Interrupt nesting behaviour configuration:

LIBRARY_LOWEST_INTERRUPT_PRIORITY	15
LIBRARY_MAX_SYSCALL_INTERRUPT_PRIORITY	5

Added with 10.2.1 support:

MESSAGE_BUFFER_LENGTH_TYPE	size_t
USE_POSIX_ERRNO	Disabled

CMSIS-RTOS V2 flags:

USE_OS2_THREAD_SUSPEND_RESUME	Enabled
USE_OS2_THREAD_ENUMERATE	Enabled
USE_OS2_EVENTFLAGS_FROM_ISR	Enabled
USE_OS2_THREAD_FLAGS	Enabled
USE_OS2_TIMER	Enabled
USE_OS2_MUTEX	Enabled

3.13.2. Include parameters:

Include definitions:

vTaskPrioritySet	Enabled
uxTaskPriorityGet	Enabled
vTaskDelete	Enabled
vTaskCleanUpResources	Disabled
vTaskSuspend	Enabled
vTaskDelayUntil	Enabled
vTaskDelay	Enabled
xTaskGetSchedulerState	Enabled

xTaskResumeFromISR	Enabled
xQueueGetMutexHolder	Enabled
xSemaphoreGetMutexHolder	Disabled
pcTaskGetTaskName	Disabled
uxTaskGetStackHighWaterMark	Enabled
xTaskGetCurrentTaskHandle	Enabled
eTaskGetState	Enabled
xEventGroupSetBitFromISR	Disabled
xTimerPendFunctionCall	Enabled
xTaskAbortDelay	Disabled
xTaskGetHandle	Disabled
uxTaskGetStackHighWaterMark2	Disabled

3.13.3. Advanced settings:

Newlib settings (see parameter description first):

USE_NEWLIB_REENTRANT Disabled

Project settings (see parameter description first):

Use FW pack heap file Enabled

3.14. USB_DEVICE

Class For FS IP: Communication Device Class (Virtual Port Com)

3.14.1. Parameter Settings:

Basic Parameters:

USBD_MAX_NUM_INTERFACES (Maximum number of supported interfaces)	1
USBD_MAX_NUM_CONFIGURATION (Maximum number of supported configuration)	1
USBD_MAX_STR_DESC_SIZ (Maximum size for the string descriptors)	512
USBD_SELF_POWERED (Enabled self power)	Enabled
USBD_DEBUG_LEVEL (USBD Debug Level)	0: No debug message
USBD_LPM_ENABLED (Link Power Management)	1: Link Power Management supported

Class Parameters:

USB CDC Rx Buffer Size	2048
USB CDC Tx Buffer Size	2048

3.14.2. Device Descriptor:

Device Descriptor:

VID (Vendor Identifier)	1155
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LANGID_STRING (Language Identifier)	English(United States)
MANUFACTURER_STRING (Manufacturer Identifier)	STMicroelectronics
Device Descriptor FS:	
PID (Product Identifier)	22336
PRODUCT_STRING (Product Identifier)	STM32 Virtual ComPort
CONFIGURATION_STRING (Configuration Identifier)	CDC Config
INTERFACE_STRING (Interface Identifier)	CDC Interface

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
ADC1	PA2	ADC1_IN2	Analog mode	No pull-up and no pull-down	n/a	LIGHT_IN_1
	PA3	ADC1_IN3	Analog mode	No pull-up and no pull-down	n/a	LIGHT_IN_2
	PA4	ADC1_IN4	Analog mode	No pull-up and no pull-down	n/a	GAUGE_IN
	PA5	ADC1_IN5	Analog mode	No pull-up and no pull-down	n/a	UX_POT
I2C2	PB10	I2C2_SCL	Alternate Function Open Drain	No pull-up and no pull-down	Very High *	
	PB11	I2C2_SDA	Alternate Function Open Drain	No pull-up and no pull-down	Very High *	
	PB12	I2C2_SMBA	Alternate Function Open Drain	No pull-up and no pull-down	Very High *	
RCC	PH0-OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	MCO [STM32F103CBT6_PA8]
	PH1-OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
SPI1	PA6	SPI1_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PB3	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
	PB5	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	TMS
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	TCK
TIM1	PE9	TIM1_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	M1_STEP
	PE11	TIM1_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	M2_STEP
	PE13	TIM1_CH3	Alternate Function Push Pull	No pull-up and no pull-down	Low	LA_PWM
TIM2	PA15	TIM2_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	UX_KNOB1_A
	PB9	TIM2_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	UX_KNOB1_B
TIM3	PA7	TIM3_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	ENC1_B
	PC6	TIM3_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	ENC1_A
TIM4	PD12	TIM4_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	ENC2_A
	PD13	TIM4_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	ENC2_B
TIM5	PA0-WKUP	TIM5_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	UX_KNOB2_A
	PA1	TIM5_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	UX_KNOB2_B
USART3	PD8	USART3_TX	Alternate Function Push Pull	No pull-up and no pull-down	Very High	STLK_RX

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
					*	[STM32F103CBT6_PA3]
	PD9	USART3_RX	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	STLK_TX [STM32F103CBT6_PA2]
USB_OTG_FS	PA8	USB_OTG_FS_SOF	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	USB_SOF [TP1]
	PA9	USB_OTG_FS_VBUS	Input mode	No pull-up and no pull-down	n/a	USB_VBUS
	PA11	USB_OTG_FS_DM	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	USB_DM
	PA12	USB_OTG_FS_DP	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	USB_DP
Single Mapped Signals	PC14-OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a	
	PC15-OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	
	PA10	USB_OTG_FS_ID	Alternate Function Push Pull	No pull-up and no pull-down	Very High *	USB_ID
GPIO	PC13	GPIO_EXTI13	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	USER_Btn [B1]
	PF0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PF1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	OLED_DC
	PF2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	OLED_RST
	PC5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LDO_EN
	PB0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LD1 [Green]
	PB1	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	BUCK12_PG
	PB2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	BUCK12_EN
	PF11	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	BUCK5_PG
	PF12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	BUCK5_EN
	PF13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M2_SLP
	PF14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M1_SLP
	PF15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M1_EN
	PG0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M1_MS2
	PG1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M1_MS1
	PE7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M1_nFLT
	PE8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M1_DIR
	PE10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M2_EN
	PE12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M2_MS2
	PE14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M2_MS1
	PE15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M2_nFLT

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PB13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	M2_DIR
	PB14	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LD3 [Red]
	PB15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LA_IN1
	PD10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LA_MODE
	PG6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	
	PG7	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	USB_OverCurrent [STMPS2151STR_FAULT]
	PG15	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	OLED_CS
	PB4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	OLED_RST
	PB6	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	OLED_DC
	PB7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LD2 [Blue]

4.2. DMA configuration

nothing configured in DMA service

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	15	0
System tick timer	true	15	0
TIM1 update interrupt and TIM10 global interrupt	true	5	0
TIM6 global interrupt and DAC1, DAC2 underrun error interrupts	true	15	0
USB On The Go FS global interrupt	true	5	0
PVD interrupt through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
ADC1, ADC2 and ADC3 interrupts	unused		
TIM1 break interrupt and TIM9 global interrupt	unused		
TIM1 trigger and commutation interrupts and TIM11 global interrupt	unused		
TIM1 capture compare interrupt	unused		
TIM2 global interrupt	unused		
TIM3 global interrupt	unused		
TIM4 global interrupt	unused		
I2C2 event interrupt	unused		
I2C2 error interrupt	unused		
SPI1 global interrupt	unused		
USART3 global interrupt	unused		
EXTI line[15:10] interrupts	unused		
TIM5 global interrupt	unused		
FPU global interrupt	unused		

4.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	false	false
Debug monitor	false	true	false
Pendable request for system service	false	false	false
System tick timer	false	false	true
TIM1 update interrupt and TIM10 global interrupt	false	true	true
TIM6 global interrupt and DAC1, DAC2 underrun error interrupts	false	true	true
USB On The Go FS global interrupt	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current

Middleware					
FREERTOS					
USB_DEVICE					
System Core	Analog	Timers	Connectivity	Multimedia	Computing
DMA	ADC1	TIM1	I2C2		
GPIO		TIM2	SPI1		
NVIC		TIM3	USART3		
RCC		TIM4	USB_FS		
SYS		TIM5			

6. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32f446_bsdI.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32f446_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32f4-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
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